

Model Outline

- One seller, N bidders
- Seller's cost is either c_L or c_H with $\Pr(c_L) = \psi$.
- Buyer's value is iid distributed according to some F supported on $[c_L, 1]$.
- 1st stage: an English auction with initially hidden reserve price, which is revealed conditional on being met
- 2nd stage: if the English auction fails, one bidder is randomly drawn to make a TIOLI offer to the seller. An individual bidder is drawn with probability $\delta \leq 1/N$.

Preliminary Analysis

- In the 2nd stage, if the chosen buyer has a belief $\tilde{\psi}$, he offers price c_H iff

$$v - c_H \geq \tilde{\psi}(v - c_L) \Leftrightarrow v \geq \frac{c_H - \tilde{\psi}c_L}{1 - \tilde{\psi}}.$$

- Consider an equilibrium with the following structure. The high-cost plays a pure-strategy r_H whereas the low-cost plays a mixed strategy denoted by $\rho \in \Delta[\underline{r}, \bar{r}]$.
- Claim to be shown: $r_H > \bar{r}$. Intuition: for any increasing bidding function, the optimal reserve price for high-cost is strictly above that of the low-cost type.
- When the ongoing bid exceeds $\bar{r}$ but it is not announced that the reserve has been met, bidders believe that the seller has high cost, so they go by $\beta(v) = v - \delta(v - c_H)$. The high-cost seller's profit is thus

$$\Pi_H(r_H) = \int_{\beta^{-1}(r_H)}^1 \int_0^v (\max\{r_H, s\} - c_H) d\left(\frac{F(s)}{F(v)}\right)^{N-1} dF(v)^N,$$

because once a bidder has $\beta(v) \geq r_H$, the reserve price is announced to have been met, so people bid according to their own value. Simplifying gives

$$\begin{aligned} \Pi_H(r_H) &= N \int_{\beta^{-1}(r_H)}^1 \left[(r_H - c_H) F(r_H)^{N-1} + \int_{r_H}^v (s - c_H) dF(s)^{N-1} \right] dF(v) \\ &= N \left[\frac{(r_H - c_H) F(r_H)^{N-1} (1 - F(\beta^{-1}(r_H))) + \int_{r_H}^1 (s - c_H) dF(s)^{N-1}}{-F(\beta^{-1}(r_H)) \int_{r_H}^{\beta^{-1}(r_H)} (s - c_H) dF(s)^{N-1} - \int_{\beta^{-1}(r_H)}^1 F(v) (v - c_H) dF(v)^{N-1}} \right]. \end{aligned}$$

- Check: The FOC seems to be

$$r_H - c_H = (1 - \delta) \frac{1 - F\left(\frac{r_H - \delta c_H}{1 - \delta}\right)}{f\left(\frac{r_H - \delta c_H}{1 - \delta}\right)} - \frac{1}{F(r_H)^{N-1}} \int_{r_H}^{\frac{r_H - \delta c_H}{1 - \delta}} (v - c_H) dF(v)^{N-1}.$$

- Recall $\rho \in \Delta[\underline{r}, \bar{r}]$ is the mixed strategy used by the low-cost seller. If the English auction fails with final bid b , then the buyer's belief is $\tilde{\psi} = \frac{\psi(1 - \rho(b))}{1 - \psi\rho(b)}$.

- Suppose a bidder with value v decides his bid in the first-stage English auction by comparing $v - b$ vs $\delta \max \left\{ v - c_H, \frac{\psi(1-\rho(b))}{1-\psi\rho(b)} (v - c_L), 0 \right\}$. His bid thus satisfies:

$$b = v - \delta \max \left\{ v - c_H, \frac{\psi(1-\rho(b))}{1-\psi\rho(b)} (v - c_L), 0 \right\}.$$

The RHS is weakly decreasing in b and bounded between $[v - \delta \max \{v - c_H, \psi(v - c_L)\}, v - \delta \max \{v - c_H, 0\}]$. The bidding strategy β given ρ is thus

$$\beta(v) = \begin{cases} v - \delta\psi(v - c_L) & \text{if } v \leq \frac{r - \delta\psi c_L}{1 - \delta\psi} \\ h^{-1}(v) & \text{if } v \in \left(\frac{r - \delta\psi c_L}{1 - \delta\psi}, v^* \right) \\ v - \delta(v - c_H) & \text{if } v \geq v^* \end{cases},$$

where

$$h(b) \equiv \frac{(1 - \psi\rho(b))b - \delta\psi(1 - \rho(b))c_L}{1 - \delta\psi - (1 - \delta)\psi\rho(b)},$$

and $v^* \in \left[\frac{r - \delta\psi c_L}{1 - \delta\psi}, \frac{c_H - \psi c_L}{1 - \psi} \right]$ is the solution to

$$v - c_H = \frac{\psi(1 - \rho((1 - \delta)v + \delta c_H))}{1 - \psi\rho((1 - \delta)v + \delta c_H)} (v - c_L) \Leftrightarrow \rho((1 - \delta)v + \delta c_H) = \frac{\psi(v - c_L) - (v - c_H)}{\psi(c_H - c_L)}.$$

- As noted above, it suffices to focus on the low-type seller's profit. Suppose it sets reserve price at $\underline{r}$. Conditional on the auction fails, which happens with probability $\Pr(\beta(v) \leq \underline{r})^N = \Pr(v \leq \beta^{-1}(\underline{r}))^N = F(\beta^{-1}(\underline{r}))^N = F\left(\frac{r - \delta\psi c_L}{1 - \delta\psi}\right)^N$, the profit is

$$\begin{aligned} & \delta N(c_H - c_L) \times \Pr\left(v_1 \geq \frac{c_H - \psi c_L}{1 - \psi} \middle| v_{\max} \leq \beta^{-1}(\underline{r})\right) \\ &= \delta N(c_H - c_L) \times \frac{\Pr\left(v_1 \geq \frac{c_H - \psi c_L}{1 - \psi}, v_{\max} \leq \beta^{-1}(\underline{r})\right)}{\Pr(v_{\max} \leq \beta^{-1}(\underline{r}))} \\ &= \delta N(c_H - c_L) \times \frac{\left[F(\beta^{-1}(\underline{r})) - F\left(\frac{c_H - \psi c_L}{1 - \psi}\right)\right] F\left(\frac{r - \delta\psi c_L}{1 - \delta\psi}\right)^{N-1}}{F\left(\frac{r - \delta\psi c_L}{1 - \delta\psi}\right)^N} \\ &= \delta N(c_H - c_L) \times \left(1 - \frac{F\left(\frac{c_H - \psi c_L}{1 - \psi}\right)}{F\left(\frac{r - \delta\psi c_L}{1 - \delta\psi}\right)}\right). \end{aligned}$$

Conditional on the auction succeeds, which happens with probability $1 - F\left(\frac{r - \delta\psi c_L}{1 - \delta\psi}\right)^N$, the profit is

$$\begin{aligned} & \int_{\beta^{-1}(\underline{r})}^1 \int_{c_L}^v (\max\{\underline{r}, s\} - c_L) d\left(\frac{F(s)}{F(v)}\right)^{N-1} dF(v)^N \times \frac{1}{1 - F\left(\frac{r - \delta\psi c_L}{1 - \delta\psi}\right)^N} \\ &= N \int_{\beta^{-1}(\underline{r})}^1 \int_{c_L}^v (\max\{\underline{r}, s\} - c_L) dF(s)^{N-1} dF(v) \times \frac{1}{1 - F\left(\frac{r - \delta\psi c_L}{1 - \delta\psi}\right)^N} \\ &= N \int_{\beta^{-1}(\underline{r})}^1 \left[(\underline{r} - c_L) F(\underline{r})^{N-1} + \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} \right] dF(v) \times \frac{1}{1 - F\left(\frac{r - \delta\psi c_L}{1 - \delta\psi}\right)^N} \\ &= N \left\{ (\underline{r} - c_L) F(\underline{r})^{N-1} (1 - F(\beta^{-1}(\underline{r}))) + \int_{\beta^{-1}(\underline{r})}^1 \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \right\} \times \frac{1}{1 - F\left(\frac{r - \delta\psi c_L}{1 - \delta\psi}\right)^N}. \end{aligned}$$

Taken together, the low-type's profit is thus

$$\begin{aligned}\Pi(\underline{r}) \equiv & \delta N (c_H - c_L) \times \left(F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) - F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) \right) F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right)^{N-1} \\ & + N \left\{ (\underline{r} - c_L) F(\underline{r})^{N-1} (1 - F(\beta^{-1}(\underline{r}))) + \int_{\beta^{-1}(\underline{r})}^1 \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \right\}.\end{aligned}$$

- Claim to be verified: The equilibrium value of $\underline{r}$ must be the argmax of $\Pi(\cdot)$ defined above. Denote it by $\underline{r}^*$.
 - Denote the argmax of $\Pi(\cdot)$ by $\underline{r}^*$. If $\underline{r} > \underline{r}^*$, deviating from $\underline{r}$ to $\underline{r}^*$ gives $\Pi(\underline{r}^*) > \Pi(\underline{r})$ and is thus profitable. If $\underline{r} < \underline{r}^*$, then deviating from $\underline{r}$ to $\underline{r}^*$ is again profitable as $\Pi(\cdot)$ underestimates the gain (as it does not take into account the upward revision of the bidder's belief about the seller's cost being high).
- The low-cost seller's eqm payoff is thus $\Pi(\underline{r}^*)$.
- Closer look at $\underline{r}^*$: $\beta^{-1}(\underline{r}) = (\underline{r} - \delta\psi c_L) / (1 - \delta\psi)$. Let's work out the FOC for maximizing $\Pi(\underline{r})$.
 - The derivative of the first line is:

$$\begin{aligned}& \frac{1}{1 - \delta\psi} f\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right)^{N-1} \\ & + (N-1) \frac{1}{1 - \delta\psi} \left(F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) - F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) \right) F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right)^{N-2} f\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) \\ & = \frac{1}{1 - \delta\psi} f\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right)^{N-2} \left[NF\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right) - (N-1) F\left(\frac{c_H - \psi c_L}{1 - \psi}\right) \right]\end{aligned}$$

- The double integral in the second line can be simplified into

$$\begin{aligned}& \int_{\beta^{-1}(\underline{r})}^1 \int_{\underline{r}}^v (s - c) dF(s)^{N-1} dF(v) \\ & = \left[F(v) \int_{\underline{r}}^v (s - c) dF(s)^{N-1} \right]_{\beta^{-1}(\underline{r})}^1 - \int_{\beta^{-1}(\underline{r})}^1 F(v) (v - c) dF(v)^{N-1} \\ & = \int_{\underline{r}}^1 (s - c) dF(s)^{N-1} - F(\beta^{-1}(\underline{r})) \int_{\underline{r}}^{\beta^{-1}(\underline{r})} (s - c) dF(s)^{N-1} - \int_{\beta^{-1}(\underline{r})}^1 F(v) (v - c) dF(v)^{N-1}\end{aligned}$$

- * The derivative of the first term reads:

$$\frac{d}{dr} \int_{\underline{r}}^1 (s - c) dF(s)^{N-1} = -(N-1) (r - c) F(r)^{N-2} f(r)$$

- * The derivative of the second term reads:

$$\begin{aligned}& \frac{d}{dr} \left(F(\beta^{-1}(\underline{r})) \int_{\underline{r}}^{\beta^{-1}(\underline{r})} (s - c) dF(s)^{N-1} \right) \\ & = f(\beta^{-1}(\underline{r})) \frac{d\beta^{-1}(\underline{r})}{dr} \int_{\underline{r}}^{\beta^{-1}(\underline{r})} (s - c) dF(s)^{N-1} \\ & \quad + (N-1) F(\beta^{-1}(\underline{r})) \left[(\beta^{-1}(\underline{r}) - c) F(\beta^{-1}(\underline{r}))^{N-2} f(\beta^{-1}(\underline{r})) \frac{d\beta^{-1}(\underline{r})}{dr} - (r - c) F(r)^{N-2} f(r) \right].\end{aligned}$$

* The derivative of the third term reads:

$$\frac{d}{dr} \int_{\beta^{-1}(r)}^1 F(v) (v - c) dF(v)^{N-1} = -\frac{d\beta^{-1}(r)}{dr} (N-1) (\beta^{-1}(r) - c) F(\beta^{-1}(r))^{N-1} f(\beta^{-1}(r)).$$

* The derivative of the double integral is thus

$$-(N-1)(r-c)F(r)^{N-2}f(r)(1-F(\beta^{-1}(r))) - f(\beta^{-1}(r)) \frac{d\beta^{-1}(r)}{dr} \int_r^{\beta^{-1}(r)} (s-c) dF(s)^{N-1}.$$

– The derivative of the first term in the bracket is

$$\begin{aligned} & \frac{d}{dr} \left[(\underline{r} - c_L) F(\underline{r})^{N-1} (1 - F(\beta^{-1}(\underline{r}))) \right] \\ &= \left[F(r) (1 - F(\beta^{-1}(r))) + (N-1)(r-c) (1 - F(\beta^{-1}(r))) f(r) \right. \\ & \quad \left. - (r-c) F(r) f(\beta^{-1}(r)) \frac{1}{1-\delta\psi} \right] F(r)^{N-2}. \end{aligned}$$

– Taken together, the FOC for maximizing $\Pi(\underline{r})$:

$$\begin{aligned} 0 &= \delta N (c_H - c_L) \frac{1}{1-\delta\psi} f\left(\frac{\underline{r} - \delta\psi c_L}{1-\delta\psi}\right) F\left(\frac{\underline{r} - \delta\psi c_L}{1-\delta\psi}\right)^{N-2} \\ & \quad \times \left[NF\left(\frac{\underline{r} - \delta\psi c_L}{1-\delta\psi}\right) - (N-1) F\left(\frac{c_H - \psi c_L}{1-\psi}\right) \right] \\ & \quad + N \left[F(r) (1 - F(\beta^{-1}(r))) + (N-1)(r-c) (1 - F(\beta^{-1}(r))) f(r) \right. \\ & \quad \left. - (r-c) F(r) f(\beta^{-1}(r)) \frac{1}{1-\delta\psi} \right] F(r)^{N-2} \\ & \quad + N \left[-(N-1)(r-c) F(r)^{N-2} f(r) (1 - F(\beta^{-1}(r))) - f(\beta^{-1}(r)) \frac{d\beta^{-1}(r)}{dr} \int_r^{\beta^{-1}(r)} (s-c) dF(s)^{N-1} \right] \end{aligned}$$

– Upon simplification,

$$\begin{aligned} \underline{r}^* - c_L &= (1 - \delta\psi) \left(\frac{1 - F(\beta^{-1}(\underline{r}^*))}{f(\beta^{-1}(\underline{r}^*))} \right) \\ & \quad + \frac{1}{F(\underline{r}^*)^{N-1}} \left[\delta (c_H - c_L) F(\beta^{-1}(r))^{N-2} \left[NF(\beta^{-1}(r)) - (N-1) F\left(\frac{c_H - \psi c_L}{1-\psi}\right) \right] \right. \\ & \quad \left. - \int_{\underline{r}^*}^{\beta^{-1}(\underline{r}^*)} (s - c_L) dF(s)^{N-1} \right] \end{aligned}$$

• The seller's profit of setting a reserve price r above $\underline{r}$:

$$\begin{aligned} \hat{\Pi}(r) &= \delta N (c_H - c_L) \\ & \quad \times \left[\left(1 - \frac{F\left(\frac{c_H - \psi c_L}{1-\psi}\right)}{F\left(\frac{\underline{r} - \delta\psi c_L}{1-\delta\psi}\right)} \right) F\left(\frac{\underline{r} - \delta\psi c_L}{1-\delta\psi}\right)^N + \int_{\frac{\underline{r} - \delta\psi c_L}{1-\delta\psi}}^{\beta^{-1}(r)} \left(1 - \frac{F\left(\frac{(1-\psi\rho(\beta(v)))c_H - \psi(1-\rho(\beta(v)))c_L}{1-\psi}\right)}{F(v)} \right) dF(v)^N \right] \\ & \quad + N \left\{ (r - c_L) F(r)^{N-1} (1 - F(\beta^{-1}(r))) + \int_{\beta^{-1}(r)}^1 \int_r^v (s - c_L) dF(s)^{N-1} dF(v) \right\} \\ &= \delta N (c_H - c_L) \\ & \quad \times \left[\left(F\left(\frac{\underline{r} - \delta\psi c_L}{1-\delta\psi}\right) - F\left(\frac{c_H - \psi c_L}{1-\psi}\right) \right) F\left(\frac{\underline{r} - \delta\psi c_L}{1-\delta\psi}\right)^{N-1} \right. \\ & \quad \left. + N \int_{\frac{\underline{r} - \delta\psi c_L}{1-\delta\psi}}^{\beta^{-1}(r)} \left(F(v) - F\left(\frac{(1-\psi\rho(\beta(v)))c_H - \psi(1-\rho(\beta(v)))c_L}{1-\psi}\right) \right) F(v)^{N-2} dF(v) \right] \\ & \quad + N \left\{ (r - c_L) F(r)^{N-1} (1 - F(\beta^{-1}(r))) + \int_r^1 (v - c_L) dF(v)^{N-1} \right. \\ & \quad \left. - F(\beta^{-1}(r)) \int_r^{\beta^{-1}(r)} (v - c_L) dF(v)^{N-1} - \int_{\beta^{-1}(r)}^1 F(v) (v - c_L) dF(v)^{N-1} \right\} \end{aligned}$$

In eqm, $\hat{\Pi}(r) = \Pi(\underline{r}^*)$ for all $r \in [\underline{r}, \bar{r}]$, i.e.,

$$\begin{aligned} & \delta N(c_H - c_L) \int_{\frac{r - \delta \psi c_L}{1 - \delta \psi}}^{\beta^{-1}(r)} \left(F(v) - F\left(\frac{(1 - \psi \rho(\beta(v))) c_H - \psi(1 - \rho(\beta(v))) c_L}{1 - \psi}\right) \right) F(v)^{N-2} dF(v) \\ & + (r - c_L) F(r)^{N-1} (1 - F(\beta^{-1}(r))) + \int_r^1 (v - c_L) dF(v)^{N-1} \\ & - F(\beta^{-1}(r)) \int_r^{\beta^{-1}(r)} (v - c_L) dF(v)^{N-1} - \int_{\beta^{-1}(r)}^1 F(v) (v - c_L) dF(v)^{N-1} \\ = & (\underline{r} - c_L) F(\underline{r})^{N-1} (1 - F(\beta^{-1}(\underline{r}))) + \int_{\beta^{-1}(\underline{r})}^1 \int_{\underline{r}}^v (s - c_L) dF(s)^{N-1} dF(v) \end{aligned}$$

- Assuming the LHS is differentiable, it looks like we have

$$\begin{aligned} 0 = & \delta N(c_H - c_L) \left(F(\beta^{-1}(r)) - F\left(\frac{(1 - \psi \rho(r)) c_H - \psi(1 - \rho(r)) c_L}{1 - \psi}\right) \right) F(\beta^{-1}(r))^{N-2} f(\beta^{-1}(r)) \frac{d\beta^{-1}(r)}{dr} \\ & + F(r)^{N-1} (1 - F(\beta^{-1}(r))) - \frac{d\beta^{-1}(r)}{dr} f(\beta^{-1}(r)) \left[(\beta^{-1}(r) - c_L) F(\beta^{-1}(r))^{N-1} - \int_r^{\beta^{-1}(r)} F(v)^{N-1} dv \right] \end{aligned}$$

- For r close to $\underline{r}$, $\beta^{-1}(r) = h(r) = \frac{(1 - \psi \rho(r)) r - \delta \psi (1 - \rho(r)) c_L}{1 - \delta \psi - (1 - \delta) \psi \rho(r)}$, and $\rho(r) = \frac{-h(r) + r + \delta \psi (h(r) - c_L)}{\psi(-h(r) + r + \delta(h(r) - c_L))}$, so the equation above becomes:

$$\begin{aligned} h'(r) = & \frac{F(r)^{N-1} (1 - F(h(r)))}{f(h(r))} \\ & \times \left[\begin{aligned} & (h(r) - c_L) F(h(r))^{N-1} - \int_r^{h(r)} F(v)^{N-1} dv \\ & - \delta N(c_H - c_L) \left(F(h(r)) - F\left(\frac{-(h(r) - r) c_L + \delta c_H (h(r) - c_L)}{-(h(r) - r) + \delta(h(r) - c_L)}\right) \right) F(h(r))^{N-2} \end{aligned} \right]^{-1} \end{aligned}$$

giving a differential equation in $h(r)$ and hence $\rho(r)$.

- Need to show that the differential equation above admits a solution.
- For r above $\beta(v^*)$, $\beta^{-1}(r) = \frac{r - \delta c_H}{1 - \delta}$, so the equation becomes

$$\begin{aligned} 0 = & \delta N(c_H - c_L) \left(F\left(\frac{r - \delta c_H}{1 - \delta}\right) - F\left(\frac{(1 - \psi \rho(r)) c_H - \psi(1 - \rho(r)) c_L}{1 - \psi}\right) \right) F\left(\frac{r - \delta c_H}{1 - \delta}\right)^{N-2} f\left(\frac{r - \delta c_H}{1 - \delta}\right) \frac{1}{1 - \delta} \\ & + F(r)^{N-1} \left(1 - F\left(\frac{r - \delta c_H}{1 - \delta}\right) \right) \\ & - \frac{1}{1 - \delta} f\left(\frac{r - \delta c_H}{1 - \delta}\right) \left[\left(\frac{r - \delta c_H}{1 - \delta} - c_L \right) F\left(\frac{r - \delta c_H}{1 - \delta}\right)^{N-1} - \int_r^{\frac{r - \delta c_H}{1 - \delta}} F(v)^{N-1} dv \right], \end{aligned}$$

giving again an equation in $\rho(r)$.

- Under the most optimistic belief for the low-cost seller, the bidder with $v \geq c_H$ offers c_H in the 2nd stage for sure, yielding an expected payoff of $\delta N c_H (1 - F(c_H)) < c_H$. Therefore, $\bar{r}$ is necessarily below that of the high-cost seller, r_H .
- Case of public reserve price. If the equilibrium is separating, the low-cost sets a reserve price given by

$$r_L - c_L = (1 - \delta) \frac{1 - F\left(\frac{r_L - \delta c_L}{1 - \delta}\right)}{f\left(\frac{r_L - \delta c_L}{1 - \delta}\right)} - \frac{1}{F(r_L)^{N-1}} \int_{r_L}^{\frac{r_L - \delta c_L}{1 - \delta}} (v - c_L) dF(v)^{N-1}.$$

whereas high-cost sets reserve price weakly above r_H , solution to

$$r_H - c_H = (1 - \delta) \frac{1 - F\left(\frac{r_H - \delta c_H}{1 - \delta}\right)}{f\left(\frac{r_H - \delta c_H}{1 - \delta}\right)} - \frac{1}{F(r_H)^{N-1}} \int_{r_H}^{\frac{r_H - \delta c_H}{1 - \delta}} (v - c_H) dF(v)^{N-1}.$$

that is enough to deter low-type's from mimicking. (There may be semi-separating eqm, etc.) Need to consider these eqm payoffs.